

# Autonomous vehicle with communicative driving for pedestrian crossing: Trajectory optimization

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**Abstract**—Connected and autonomous vehicles (CAV) is a key technology for this century. One of the main challenges is to define a smart interaction behavior of CAV with the other road users. The challenge is mainly raised at conflicting points where path of CAV intersects with the other users. Recent studies show interaction with humans is a big challenge. It not only requires a collision avoidance system but also more communicative behaviors of the CAV. More precisely, pedestrian needs to understand the intention of the incoming CAV whether it will cross first or not according to its speed profile. One way to overcome this issue is to design optimal trajectory control of the CAV that matches with the pedestrian expectation. However, such a design faces the non-linearity of the space sharing constraint. This paper uses sequence modeling based on Petri Net in order to overcome the problem. It also uses Hamiltonian analysis to derive the optimal control. Numerical examples of the communicative behavior of CAV at pedestrian crossing show that the proposed approach provides CAV with a kind of automatic courtesy.

## I. INTRODUCTION

Connected and autonomous vehicles (CAV) technology allows to improve safety and efficiency of the transportation system in terms of travel time and energy consumption [1][2]. Transportation in closed industrial sites provides a conducive environment for autonomous and connected vehicle technologies. Indeed, the scope of operation can be extended gradually, by allowing interactions with other road users. One important step to gain autonomy in industrial sites is collision avoidance with obstacles [3]. However this is not enough to behave with humans at intersection points. Indeed, they seek for a communicative behavior in order to dynamically interact with CAV and find a common optimal solution. This needs techniques that optimise interaction between road users and CAV. Many researches have been done recently to answer this issue as presented in [4][5][6].

A part of the solution to this problem is a signalization system at the CAV level as showed in Fig. 1. The CAV emits a signal that can be easily interpreted by pedestrians, such as a pictogram displayed on the CAV's grill and/or projected on the road near the pedestrian. This informs the pedestrian first that she/he is detected by the CAV and second, about the intention of the vehicle. This allows an explicit communication with the pedestrian. The other part of the solution lies in the decision process that determines

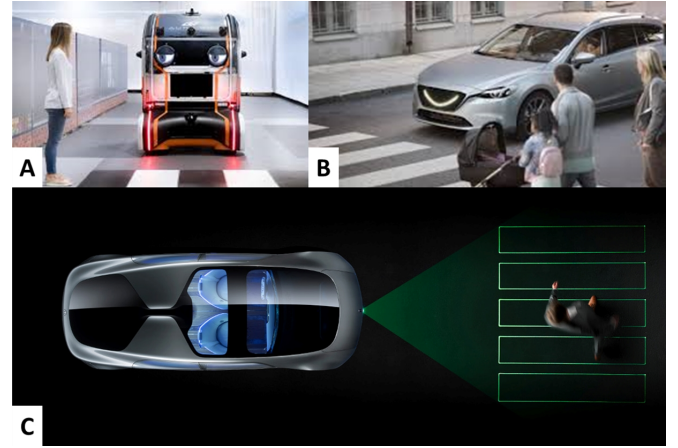


Fig. 1. Signalisation system at the CAV level: A- The CAV informs the pedestrian that she is detected -B A smile pictogram displayed at the CAV's grill -C A pictogram projected on the road

who goes first and accordingly computes the suitable speed profile of the CAV.

The speed profile is not neutral [7], [8]. It should provide a communicative behavior for confirming the intention of the CAV. The speed profile is the implicit communication that should not contradict the signalization, i.e, explicit communication. For instance, a CAV that displays the green but accelerates to resort at the last seconds to the emergency brake is not a suitable behavior.

To design the speed profile, this paper considers a CAV which interacts with the pedestrian in order to optimise with her/him a given objective function. The choice of the function must fit the pedestrian behavior. Unfortunately, studying the suitable function faces many challenges. The most important one is to overcome the non-linearity of the occupancy constraint that prevents from getting an explicit solution.

Such an approach is recently investigated for cooperative CAV at intersections. Because, the studied systems are limited to robots, it is hard to extend the results directly to the human-robot interaction. Many works avoid the non-linearity by studying a distributed problem where the sequence and consensus times are predefined [9]. However, this prevents from addressing the consensus issue. One notable work is presented in [10]. The authors developed a distributed coordinated algorithm to obtain real-time CAV trajectories. The algorithm can only be applied to find a consensus between CAVs since the speed profile is computed by relaxing the conflict constraint. To overcome the non-linearity,

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[11] formulates the problem in space coordinates, rather than time. This considerably simplifies the conflict constraint formulation. However, because of using space coordinates, the state variable  $z(p) = 1/v(p)$  (with  $p$  and  $v$  being the position and the speed, respectively), prevents from studying cases when the speed equals zero.

To provide a solution to the modeling problem of the conflicting space, this paper considers this space as a resource shared system between the CAV and the pedestrian. The access to the resource and the exit are modeled by a Petri Net (PN). This allows to obtain a suitable formulation of the system. The conflict model is used to derive the optimisation problem and to obtain the optimal trajectory for each part (i.e. the CAV and the pedestrian) by using a Hamiltonian analysis. The speed profile is discussed through a numerical example.

This paper is organised as follows. First, it introduces the studied conflict problem in section II. The section III gives the mathematical model of the conflicting space. The section IV applies the Hamiltonian analysis to the problem. A numerical example (section V) is presented and discussed according to test data.

## II. PROBLEM STATEMENT

Let us consider a CAV that is coming at its desired speed ( $\tilde{v}_v$ ), and also a pedestrian who wants to cross the road. As the vehicle, the pedestrian also has a desired speed ( $\tilde{v}_p$ ). Therefore, according to their desired speed, the Conflict Zone (CZ) needs to be shared by the pedestrian and the CAV, as shown in Fig. 2. The shared space (CZ) is the colored rectangle. Both must adjust their speed to avoid collision. The dynamic of both is formulated as follows:

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} X(t) + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} U(t) \quad (1)$$

Such that in (1),  $X^T(t) = [d_v(t), v_v(t), d_p(t), v_p(t)]$  and  $U^T(t) = [u_v(t), u_p(t)]$ , with  $d_k(t)$ ,  $v_k(t)$  and  $u_k(t)$  being the traveled distance, the speed and the acceleration of the agent  $k \in \{v, p\}$ , respectively. When  $k = v$ , it denotes the vehicle whereas if  $k = p$ , it denotes the pedestrian.

Speed and acceleration constraints are considered as follows:

$$u_k(t) - \bar{u}_k \leq 0 \quad (2)$$

$$\underline{u}_k - u_k(t) \leq 0 \quad (3)$$

$$v_k(t) - \bar{v}_k \leq 0 \quad (4)$$

$$0 - v_k(t) \leq 0 \quad (5)$$

$\bar{u}_k$  and  $\underline{u}_k$  designate the maximum and the minimum acceleration (deceleration), respectively. The maximum speed is  $\bar{v}_k$ . Equation (5) avoids both pedestrian and CAV to move back but allows them to stop.  $\bar{u}_k$ ,  $\underline{u}_k$  and  $\bar{v}_k$  are defined not only according to physical limitation but also by taking into account criteria such as the safety and convenience

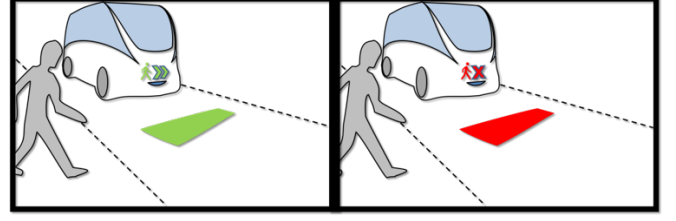


Fig. 2. CAV interacting with pedestrian for sharing the conflict space.

of passengers and good on the autonomous shuttle and maintenance costs.

To be able to extract a communicative behaviour, it remains to define:

- Objective function: The function must match as well as possible the behavior and the understanding of the pedestrian when she/he crosses the road.
- Conflict constraint: Both CAV and pedestrian must be able to have an exclusive access to the conflict space.

In the following, the paper will focus on the constraint to share safely the conflict space. The constraint should allow an explicit analysis of the speed profile in order to test different objective functions and deduce the suitable one that formulates the logic behind pedestrian behaviour.

Without loss of generality, the paper considers the minimization of the following objective function:

$$J(u_v, u_p) = \int_0^{t_f} (v_v(t) - \tilde{v}_v)^2 dt + w_p \int_0^{t_f} (v_p(t) - \tilde{v}_p)^2 dt \quad (6)$$

$w_p \geq 1$  is the weighting factor of the pedestrian and  $t_f$  is the time horizon of optimization.  $w_p \in [1, +\infty]$  acts as cursor to give more or less advantage to pedestrian according to the policy of the industrial site.  $t_f$  is defined to allow both agents to regain their desired speed after they cross CZ.  $t_f = \max(t_{v,f}, t_{p,f})$  with  $t_{k,f}$  is the time needed for an agent  $k$  to recover  $\tilde{v}_p$ . One can note from the definition of the objective function in (6), that  $J(u_v, u_p)$  penalizes the deviation from the desired speed of each user of CZ during  $t_f$ .  $t_f$  is computed according to the speed profile presented in section IV. In the following, we solve the problem in general case without giving values to neither to  $t_f$  nor to  $w_p$ . A discussion about the value  $w_p$  will be provided in section VI, whereas  $t_{k,f}$  can be easily computed from the optimal trajectory.

## III. CONFLICT MODELING

For the sake of readability, in this section we only consider the conflict between a pair of pedestrian and CAV. Fig. III-A shows the studied system. Since we only consider the conflict between a pair of a pedestrian and a CAV, then Fig. III gives the corresponding P-Timed PN model (TPN) of the conflict space. Each transition of the TPN model  $x_{k,i}$  with  $i \in \{enter, exit\}$  is associated to a counter firing function  $f_{k,i}(t)$  that gives the number of times the transition has been fired until date  $t$ . The marking of places  $p_{k,out}$  represents the state when there is a pedestrian ( $k = p$ ) or a CAV ( $k = v$ ) before the conflict zone (CZ).  $p_{k,out}$  is timed according to the

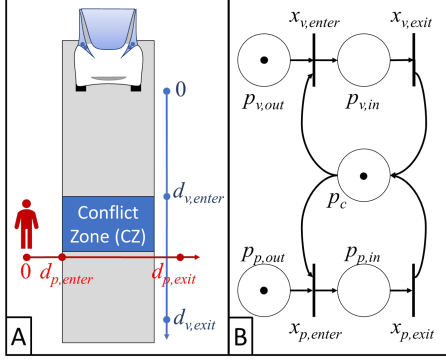


Fig. 3. Elementary pedestrian crossing system: A-System description, B-TPN model of the system.

speed of  $k$  and the remained distance to reach the conflict zone. A token in  $p_{k,in}$  means that  $k$  is occupying CZ. The associated time to  $p_{k,in}$  is the time needed by  $k$  to free CZ.  $p_c$  controls the access to CZ allowing only one agent on it.

$p_c$  is an immediate place. This means when CZ is free, each agent can immediately access to CZ. We draw the reader attention to the fact that the sojourn time associated to  $p_{k,in}$  and  $p_{k,out}$  needs to be optimized, as it is showed in the next section. Indeed, both vehicle and pedestrian adjust their speed to find a consensus time  $t_c$  for crossing CZ.

Because each place  $p_{k,out}$  owns only one token,  $f_{k,i}(t)$  equals either 0 or 1 for all  $(k,i)$ . From [12], the access to CZ is safe if the following constraint holds:

$$f_{p,enter}(t) - f_{p,exit}(t) + f_{v,enter}(t) - f_{v,exit}(t) - 1 \leq 0 \quad (7)$$

In order to associate the counter functions  $f_{k,i}$  with the dynamic of both pedestrian and CAV, the minimum traveled distances to fire transitions  $x_{k,i}$  from the origin (0) is defined as showed in Fig. III-A.  $d_{exit,k}$  includes the length of  $k$  in such a way the CZ is totally cleared. Hence, we have:

$$f_{k,i}(t) = \begin{cases} 0 & \text{if } d_k(t) < d_{k,i} \\ 1 & \text{if } d_k(t) \geq d_{k,i} \end{cases} \quad (8)$$

According to (8),  $f_{k,i}$  is a discrete function that needs to be approximated by a continuous function. For this purpose, we consider the following approximation:

$$f_{k,i}(t) \simeq fs_{k,i}(t) = \frac{1}{1 + e^{\lambda(d_{k,i} - d_k(t))}} \quad (9)$$

$fs_{k,i}$  have the following derivative function:

$$\frac{\partial fs_{k,i}}{\partial d_k} = \lambda fs_{k,i}(t) (1 - fs_{k,i}(t)) \quad (10)$$

In (9), as  $\lambda$  get bigger, the sigmoid function  $fs_{k,i}$  get closer to the discrete behavior of  $f_{k,i}$ . In order to keep CZ without overlapping movement when using  $fs_{k,i}$ , a safety distance  $sd$  is included according to the value of  $\lambda$ . In the following we assume that  $\lambda$  has a very large value. It results from (10):

$$\lim_{\lambda \rightarrow \infty} \frac{\partial fs_{k,i}}{\partial d_k} \rightarrow \begin{cases} 0 & \text{if } d_k(t) \notin [d_{k,i} - ds, d_{k,i} + ds] \\ & \text{with } ds \rightarrow 0 \\ +\infty & \text{if } d_k(t) \rightarrow d_{k,i} \end{cases} \quad (11)$$

Indeed, if  $d_k = d_{k,i}$  then  $\frac{\partial fs_{k,i}}{\partial d_k} = \frac{1}{4}\lambda$ . In the following,  $fs_{k,i}$  with a very large  $\lambda$  is used instead of  $f_{k,i}$ . It remains to provide the sojourn time to each token representing the agent  $k$  in places  $p_{k,out}$  and  $p_{k,in}$ , by optimizing the speed profile of each agent.

#### IV. OPTIMAL TRAJECTORY

In order to obtain the optimal trajectory analytically for real time applications, we use a Hamiltonian analysis of the problem. In this analysis, we consider that when the CAV detects a pedestrian as it enters the control zone. We consider here that both CAV and pedestrian cooperates to minimize  $J(u_v, u_p)$ . According to the dynamic equations (1) and the constraints of accelerations (2 and 3) and of speeds (4 and 5), the Hamiltonian function is formulated as follows:

$$\begin{aligned} H(t, x(t), u(t)) = & (v_v(t) - \tilde{v}_v)^2 + w_p(v_p(t) - \tilde{v}_p)^2 \\ & + \sum_{k \in \{v, p\}} \left( \lambda_k^d v_k(t) + \lambda_k^v u_k(t) \right. \\ & + \mu_{1,k}(u_k(t) - \bar{u}_k) + \mu_{2,k}(\underline{u}_k - u_k(t)) \\ & + \mu_{3,k}(v_k(t) - \bar{v}_k) + \mu_{4,k}(0 - v_k(t)) \left. \right) \\ & + \mu_5 \left( \sum_{k \in \{v, p\}} (fs_{k,enter}(t) - fs_{k,exit}(t)) - 1 \right) \end{aligned} \quad (12)$$

Where  $\lambda_k^d$  and  $\lambda_k^v$  are the costates and  $\mu_{1,k}$ ,  $\mu_{2,k}$ ,  $\mu_{3,k}$ ,  $\mu_{4,k}$  and  $\mu_5$  are the Lagrange multipliers. with :

$$\mu_{1,k} = \begin{cases} > 0 & \text{if } u_k(t) = \bar{u}_k \\ 0 & \text{if } u_k(t) \neq \bar{u}_k \end{cases} \quad (13)$$

$$\mu_{2,k} = \begin{cases} > 0 & \text{if } u_k(t) = \underline{u}_k \\ 0 & \text{if } u_k(t) \neq \underline{u}_k \end{cases} \quad (14)$$

$$\mu_{3,k} = \begin{cases} > 0 & \text{if } v_k(t) = \bar{v}_k \\ 0 & \text{if } v_k(t) \neq \bar{v}_k \end{cases} \quad (15)$$

$$\mu_{4,k} = \begin{cases} > 0 & \text{if } v_k(t) = 0 \\ 0 & \text{if } v_k(t) \neq 0 \end{cases} \quad (16)$$

$$\mu_5 = \begin{cases} > 0 & \text{if an agent } k \text{ is in CZ} \\ 0 & \text{if CZ is empty} \end{cases} \quad (17)$$

The Euler-Lagrange equation  $\frac{\partial H}{\partial x} = \dot{\lambda}$  becomes:

$$\begin{aligned} \dot{\lambda}_k^d &= \frac{\partial H}{\partial d_k} \\ &= \mu_5(t) \lambda \left( fs_{k,enter}(t) (1 - fs_{k,enter}(t)) \right. \\ &\quad \left. - \lambda fs_{k,exit}(t) (1 - fs_{k,exit}(t)) \right) \end{aligned} \quad (18)$$

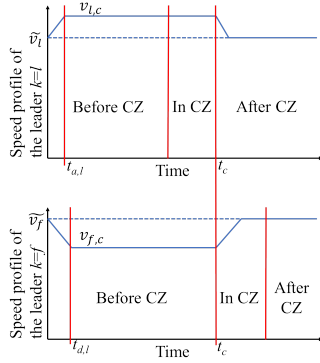


Fig. 4. Optimal speed profiles of the leader and the follower

and:

$$\lambda_v^k = \frac{\partial H}{\partial v_k} = 2 \cdot w_k v_k(t) - 2 \cdot w_k \tilde{v}_k + \lambda_p^k(t) + \mu_{3,k}(t) - \mu_{4,k}(t) \quad (19)$$

with  $w_v = 1$ .

From (18) and (11),  $\lambda_k^d$  is formulated as follows:

$$\lambda_d^k(t) = \begin{cases} a^{b,k} & \text{if agent } k \text{ is before CZ,} \\ a^{i,k} & \text{if agent } k \text{ is in CZ,} \\ a^{a,k} & \text{if agent } k \text{ is after CZ,} \end{cases} \quad (20)$$

with  $a^{b,k}$ ,  $a^{i,k}$  and  $a^{a,k}$  being constants. In the following, we put  $\lambda_p^k(t) = a^{j,k}$ , with  $j \in \{b, i, a\}$  and the value of  $j$  depends on if the agent  $k$  is before or after CZ.

The necessary condition of the optimality is  $\frac{\partial H}{\partial u_k} = 0$ , which gives the following result:

$$\begin{aligned} \frac{\partial H}{\partial u_k} &= \lambda_v^k(t) + \mu_{1,k}(t) - \mu_{2,k}(t) = 0 \\ \lambda_v^k(t) &= \mu_{2,k}(t) - \mu_{1,k}(t) \end{aligned} \quad (21)$$

from (21), (13) and (14),  $\lambda_v^k(t)$  is formulated as follows:

$$\lambda_v^k(t) = \begin{cases} -\mu_{1,k}(t) & \text{if } u_k(t) = \bar{u}_k \\ \mu_{2,k}(t) & \text{if } u_k(t) = \underline{u}_k \\ 0 & \text{otherwise} \end{cases} \quad (22)$$

From the above analysis, the obtained formulation of the speed profile of the agent  $k$  is as follows:

$$v_k(t) = \tilde{v}_k + \frac{-a^{j,k} - \mu_4(t) + \mu_5(t) - \dot{\mu}_{1,k}(t) + \dot{\mu}_{2,k}(t)}{2 \cdot w_k} \quad (23)$$

The analysis of (23) leads to the speed profiles depicted in Fig. 4. Indeed, (23) gives us an expression of  $v_k(t)$  that varies according to the constraints of speed and acceleration as well as to the position of the agent. Simply speaking, (23) expresses that the agent  $k$  must either keeps its speed constant or accelerate and decelerate at either  $\bar{u}_k$  or  $\underline{u}_k$ . Because of  $a^{j,k}$ , this speed variations depend on if the agent is before, inside or after the conflict zone.

The optimal speed profiles presented in Fig. 4 depends on whether or not the agent  $k$  crosses the conflict zone

first. More precisely, both agents find a consensus time  $t_c$  that minimizes  $J(u_v, u_p)$ . The leader accelerates ( $\bar{u}_k$ ) during  $t_{a,l}$  and keeps a constant high speed  $v_{l,c}$  to clear CZ at the consensus time, whereas the follower decelerates ( $\underline{u}_k$ ) during  $t_{d,f}$  and keeps a low constant speed  $v_{f,c}$  to reach CZ at the clearance time.

We notice that the curves make both agents act with a kind of ‘‘courtesy’’. The first accelerates to let the second cross sooner. Both behaviours are communicative. The leader agent accelerates and shows its intention to cross CZ first to the other. The same is true for the follower agent who decelerates, meaning that it yields the right of way to the other agent.

We draw the reader attention to the fact that the human behavior is dynamic and adaptive. Hence, a particular attention to the safety conditions requires to be addressed. Indeed, if the CAV is the follower, it must be able to stop if the pedestrian would change suddenly her/his direction in CZ. Moreover, even if the CAV is designed to respect the speed profile, experience has shown that the CAVs are not as controllable as expected ([13] and [14]). To this end, the following buffer distance  $bd$  is considered to determine a safe speed profile for the follower:

$$bd = -\frac{v_{f,c}^2}{2 \cdot \underline{u}_f} \quad (24)$$

$bd$  allows the agent to safely stop before CZ. According to  $bd$ , the speed profile is easily adapted. Indeed, from Fig. 4,  $v_{f,c} = \underline{u}_f \cdot t_{d,f} + \tilde{v}_f$ . Hence,  $t_{d,f}$  is obtained by solving the following equation:

$$\begin{aligned} \frac{1}{2} \underline{u}_f \cdot t_{d,f}^2 + \tilde{v}_f \cdot t_{d,f} + (\underline{u}_f \cdot t_{d,f} + \tilde{v}_f)(t_c - t_{d,f}) \\ = d_{f,enter} + \frac{(\underline{u}_f \cdot t_{d,f} + \tilde{v}_f)^2}{2 \cdot \underline{u}_f} \end{aligned} \quad (25)$$

In (26), the left side of the equality gives the traveled distance by the CAV until  $t_c$  whereas the right side constraints the distance to allow the agent to stop before CZ. More precisely the right side equals the remained distance to enter CZ minus the buffer distance given in (24). In this quadratic equation, there is a solution only if the  $d_{v,enter}$  is long enough to allow the CAV to stop:

$$t_{d,f} = \frac{\tilde{v}_f - \underline{u}_f t_c - \sqrt{sc}}{-2 \cdot \underline{u}_f} \quad (26)$$

with

$$\underbrace{2 \cdot \underline{u}_f \left( \frac{1}{2} \underline{u}_f t_c^2 + \tilde{v}_f t_c - 2 \cdot d_{f,enter} \right) - \tilde{v}_f^2}_{sc} \geq 0 \quad (27)$$

## V. NUMERICAL EXAMPLE

This numerical example bolsters the behaviour of the CAV by extending the conflict presented previously to 3 agents. We consider a platoon of 2 CAVs with a pedestrian who wants to cross the road as presented in Fig. 5. The connectivity of the CAV allows them to communicate together



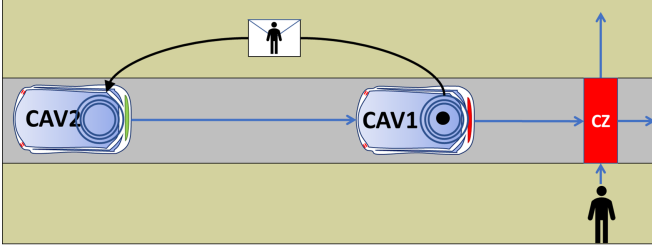


Fig. 5. Example of a platoon of 2 CAV that cooperate with the pedestrian to let her/him cross

to provide a coherent behavior to the pedestrian. Hence, as shown in Fig. 5, if CAV 1 refuses the request of the pedestrian for safety reasons, it sends a message to CAV 2. The latter will prepare itself, if possible, to allow the pedestrian to cross the road. CAV 1 communicates to CAV 2 the computed  $t_c$  that is the time it will clear CZ for letting the pedestrian cross.

TABLE I  
NUMERICAL DATA

| Data                    | CAV1 | CAV2 | Pedestrian |
|-------------------------|------|------|------------|
| $\bar{u}_k$ ( $m/s^2$ ) | 2    | 2    | 1          |
| $u_k$ ( $m/s^2$ )       | -3   | -3   | -1         |
| $\bar{v}_k$ ( $m/s$ )   | 10   | 10   | 1          |
| $\tilde{v}_k$ ( $m/s$ ) | 15   | 15   | 1.83 [15]  |
| $d_{k,enter}$ (m)       | 16   | 46   | 1.5        |
| $d_{k,exit}$ (m)        | 21   | 51   | 6          |
| $w_k$                   | 1    | 1    | 20         |

Numerical data are given in Table V. In this example, CAV 1, CAV 2 and the pedestrian arrive at their desired ( $v_k(0) = \tilde{v}_k$ ) speed. In Fig. 5, CZ is the red rectangle. The initial distance of each agent from the red rectangle is  $d_{k,enter}$ .  $d_{k,exit}$  includes the length of the occupied space by the agent. Finally, the emergency deceleration of CAV equals  $-9m/s^2$ . The two following pedestrian behaviors are considered:

- **Scenario 1:** The pedestrian behaves as expected by the speed profile
- **Scenario 2:** The pedestrian behaves as expected until she/he is near the end of CZ. She/he stops 0,1m before  $d_{p,exit}$ .

The optimal solution (scenario 1) is to let CAV 1 cross first. It frees CZ after 2.01s. The pedestrian is able then to safely access to it. She/he slowdowns first to cross when CZ is cleared. CAV 2 slowdowns too during 1s, creating a sufficient gap with CAV 1 to allow the pedestrian crossing safely the road. When CZ is cleared by CAV 1, The pedestrian speeds up to free the zone sooner (according to  $J$ ). CAV 2 moves slowly until the pedestrian leaves CZ. The trajectories of the optimal scenario are presented in Fig. 6. Red triangles indicate when the pedestrian enters to CZ and leaves CZ. One can note that both pedestrian and CAV 2 do not access immediately CZ, in order to be able to stop in the case of unforeseeable behavior of the precedent agent.

Scenario 2 aims to test the safety issue raised by the uncontrollable behavior of the pedestrian. For several reasons,

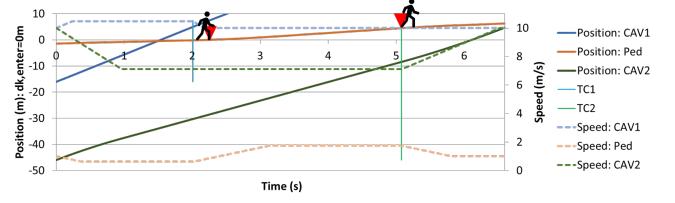


Fig. 6. **Scenario 1:** Optimal trajectories of the three agents : positions and speeds

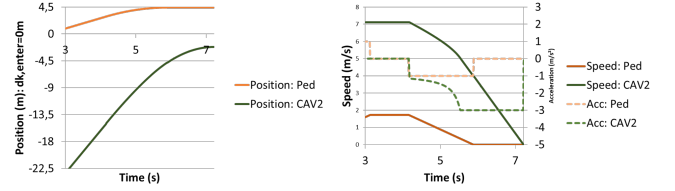


Fig. 7. **Scenario 2:** Trajectory variation according to unforeseeable behavior of the pedestrian : positions, speeds and accelerations

the pedestrian may slowdown and even to completely stop in CZ. She/he also might access to the zone later than expected because of the delayed access of CAV1. Fig. 6 shows the result under extreme conditions in which the pedestrian stops before the end of CZ. Fig. 7 shows that CAV2 immediately reacts to the pedestrian when she/he slowdowns. This figure shows also that the deceleration of CAV 2 doesn't exceed the deceleration bound  $u_{cav2}$ . CAV 2 is thus able to comfortably stop before CZ thanks to (26) and (27).

Finally, Table II compares Scenario 1 (Sc1) with a Spontaneous Behavior (SB) of each agent. In SB, each agent maintains its comfortable state ( $\tilde{v}_k$ ) whenever it is possible and the pedestrian has the priority if CAV<sub>i</sub> can stop at  $u_{cavi}$ . One can note from this table in line  $t_{k,exit}$  that the optimal trajectory allows all agent to cross CZ sooner than in in SB. This is owed to the objective function that:

- distributes the burden of the conflict over all agents: Leaders (CAV 1 and the pedestrian) speed up to let the follower (the pedestrian and CAV 2) access to CZ sooner.
- makes each follower agent maintain the appropriate speed without being necessarily resorted to a complete stop.

TABLE II  
COMPARISON BETWEEN THE OPTIMAL TRAJECTORIES AND SPONTANEOUS ONES (SB):  $t_{k,exit}$  IS THE TIME WHEN THE AGENT  $k$  LEAVES CZ, AND  $J_k = \int_0^{t_{k,f}} (v_k(t) - \tilde{v}_k)^2 dt$

| Agent             | CAV1        |             | CAV2  |               | Pedestrian        |             |
|-------------------|-------------|-------------|-------|---------------|-------------------|-------------|
| Scenarios         | Sc1         | SB          | Sc1   | SB            | Sc1               | SB          |
| $t_{k,exit}$ (s)  | 2.01        | <b>2.10</b> | 6.63  | <b>9.36</b>   | 5.08              | <b>7.03</b> |
| $t_{k,f}$ (s)     | <b>2.17</b> | 0           | 6.52  | <b>12.07</b>  | <b>5.81(2.4)</b>  | 3.11        |
| $J_k$ ( $m^2/s$ ) | <b>0.43</b> | 0           | 40.81 | <b>360.75</b> | <b>1.57(0.28)</b> | 0.78        |

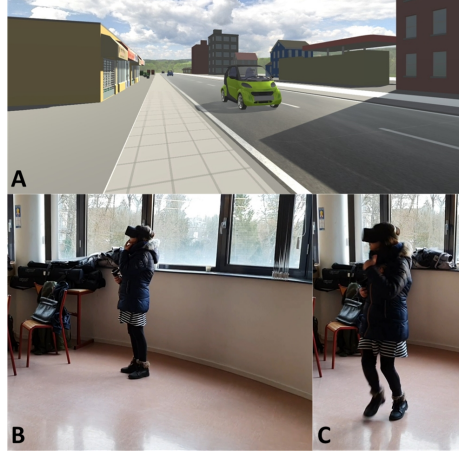


Fig. 8. Mixed reality tests of the pedestrian crossing: A-Observed virtual environment by the testers B- A tester willing to cross the intersection C- The tester quickens the pace while crossing

## VI. DISCUSSION THROUGH TESTS

We performed a series of tests using a mixed reality platform (see Fig. 8). The platform is based on the Oculus Quest, where a real tester physically crosses a virtual road by interacting with a stream of driverless vehicles. The vehicles do not emit any signal and their trajectories are not optimised. 42 virtual road crossings have been achieved by 28 testers. They had two different reactions. The first was to make vehicles dangerously slowdown ( $35,71\%$ ) with many hesitating movements. When the vehicles come to a complete stop, the pedestrian crosses. In the second behaviour ( $64,28\%$ ) the testers quicken the pace and even run, in order to avoid strong deceleration of the vehicle. Thus, we had two different deceleration ranges: less than  $-3m/s^{-2}$  and close to  $-9m/s^{-2}$ . In both cases the testers suffered from the lack of communicative behaviour.

The proposed approach in this paper seeks to provide more communicative behavior, in addition to vehicle signaling system. The numerical example shows that CAV1 and CAV2 interact with the crossing intention of the pedestrian by creating a comfortable space for crossing. The objective function allows firstly to provide a suitable behavior for both CAVs. Secondly, if the pedestrian react as expected, the crossing process allows them to free sooner the CZ (gaining  $31.15\%$ ) and to quickly recover the initial speed. In other words, the proposed approach improves the traffic efficiency at industrial sites, avoiding a complete stop.

Since our objective is to propose a CAV cooperative system for pedestrian crossing in industrial sites, we dealt with an optimization problem. The weight chosen for pedestrians can be defined to find a balance between the duty of CAVs in the industrial site and that of the crew. Many functions can be studied also in order to provide a CAV behavior that firstly fits the human understanding and secondly, benefits from an optimisation objective such as saving energy. Finally, we draw the reader attention that the same approach can be extended not only to the optimization of other objective

functions but also to the equilibrium purpose.

In order to provide a suitable objective and  $w_p$  many tests with mixed reality platform are programmed. Moreover, the approach deserves to be extended to more complex environment with several vehicles and pedestrians (batch of pedestrians). This targets a network of several CAVs that communicate together to find a common solution for a group of pedestrians who cross the road. The vehicles will be then able to provide the suitable signal to pedestrian instead of the current traffic lights.

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